Archit Shukla

San Francisco, CA | architshukla546@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Software engineer working in Python, C++, & Go on Linux, across distributed systems, Kubernetes infrastructure, & high volume Kafka event streaming. Builds the distributed backend services and cluster tooling large scale platforms run on, with security checks, regression tests, metrics, & tracing in place before launch. Contributes code to several active open source ML & AI infrastructure projects.

Education

University of Colorado Boulder • Boulder, CO — Aug 2024 – May 2026
Master of Science • Computer Science • GPA: 3.9/4
Relevant Coursework: Advanced Algorithms, Graduate Research in ML & Distributed Systems, Big Data Architecture

SRM Institute of Science & Technology • Chennai, India — Aug 2020 – May 2024
Bachelor of Technology • Computer Science & Engineering • GPA: 3.81/4
Relevant Coursework: Operating Systems, Computer Networks, Database Management Systems, Computer Architecture

Work Experience

EcoServants — Aug 2026 – Present
Software Development Intern–Backend Platform — Remote
- Owned proposal autosave with versioned MySQL revisions and HMAC signed unsubscribe tokens, rechecking state under the queue worker's lock so no draft is lost and no race slips through.
- Built backend features across an environmental nonprofit's PHP/MySQL platform with coding agents, gating every agent written patch behind mutation testing that caught **13 of 13** injected faults across 310+ automated checks.
- Validated a classifier that sorts errors into **7** categories, so retryable ones no longer fail permanently.

Peleg Lab, BioFrontiers Institute — Aug 2025 – Jan 2026
Graduate Researcher–Distributed Systems & Infrastructure — Boulder, CO
- Architected distributed AI inference pipelines in Python and multithreaded C++ across multinode Linux GPU clusters on GCP, overlapping data staging with compute to lift evaluation throughput **35%**.
- Containerized vLLM inference on Kubernetes with test gated rollouts, running **300+** sweeps with zero failures.
- Isolated concurrency faults in C++ inference workers with GDB and Valgrind, shrinking failure diagnosis time **40%**.

CodeClause — Jun 2024 – Jul 2024
Software Engineering Intern–Backend & Data Systems — Bangalore, India
- Rebuilt REST microservices and their SQL query paths behind connection pooling, index tuning, and async non blocking I/O, pulling p95 latency from **310ms to 180ms** at thousands of requests per day.
- Hardened API error handling and added Pytest regression suites, cutting bug diagnosis from **40 minutes to 26**.
- Shipped containerized deploys on Jenkins and GitHub Actions, moving release failures from **1 in 6 builds to 1 in 24**.

Adflix Media — Apr 2023 – Sep 2023
Software Engineer Intern–Cloud Infrastructure & Backend — Pune, India
- Scaled Kafka consumers and Spark ETL jobs across Dockerized AWS services, landing **5M+ records daily** into warehouse tables with offset tracking and idempotent writes that ended manual reprocessing.
- Wired Prometheus and Grafana alerts on throughput and consumer lag, holding p99 lag **under 90 seconds** at peak.
- Trimmed recovery time **27%** by routing each alert to its owning service with a linked runbook.

Technical Skills

- **Languages:** Python, C++, Go, Java, C, Bash, SQL, PHP
- **Distributed Systems:** Apache Kafka, gRPC, REST APIs, Microservices, Apache Spark, Concurrency, Fault Tolerance
- **Systems & Infrastructure:** Linux (Ubuntu, RHEL), Kubernetes, Docker, Helm, Kubebuilder, vLLM, GCP, AWS, Multithreading, CI/CD, Git
- **Data & Storage:** PostgreSQL, MySQL, Redis, Neo4j, Qdrant, ETL Pipelines, Query Optimization
- **Observability & Debugging:** Prometheus, Grafana, Pytest, GDB, Valgrind, tcpdump, Wireshark, Regression Testing

Projects

Kube-Op-Lite – GPU Workload Operator | *Go + Kubebuilder + Helm + Prometheus* [Repo] — Jun 2026 – Present
- Designed a Kubernetes operator in Go with Kubebuilder, defining a GpuWorkload CRD and an idempotent reconciliation loop that detects drift and restores failed GPU pods to desired state in **under 30 seconds** with no manual action.
- Enforced rollout safety with validating and defaulting admission webhooks plus a Helm chart, rejecting replicas under 1, empty images, and GPU counts over 8, and cut cluster bring up from **45 minutes to 8**.
- Added a BenchmarkRun CRD with GPU telemetry and baseline analysis, catching **17 of 20** injected regressions, no false alarms.

Kith – Full Stack Multi Agent AI Platform | *React + TypeScript + FastAPI + LangGraph + Redis* [Demo] — Jun 2026 – Present
- Engineered a provider agnostic LLM gateway in async FastAPI over hosted APIs and a local vLLM endpoint, with a Redis token bucket rate limiter and circuit breaker that completed **99.2%** of agent runs during simulated provider outages.
- Instrumented every LangGraph node with Langfuse tracing and an LLM judge harness, lifting extraction F1 from **0.74 to 0.91**.
- Benchmarked speculative decoding on vLLM with a small draft model against an 8B target, reaching **2.2x** decode throughput.

RepoGraph V2 – Distributed Analysis Platform | *Python + C++ + REST APIs + Linux* [Paper] — Aug 2025 – Dec 2025
- Modeled call and dependency graphs over **100K+ LOC** repositories in a distributed Python and C++ backend on Linux, localizing order dependent flaky tests caused by shared state under multiprocess load.
- Slashed inter service data transfer **70 to 85%** by sending each worker only the graph scoped slice of the repository it needed, plus a tighter serialization format between services.